CIRCULATING SUB

BALL-DROP CIRCULATION SUB

DRILLING

GENERAL DESCRIPTION

SIMPLE, RUGGED CONVENTIONAL BALL-DROP CIRCULATION SUB IN A SINGLE-COLLAR CONFIGURATION WITH NO MID-BODY CONNECTION. IT IS EASY TO USE AND OFFERS VARIABLE CYCLE CAPABILITY. A NOVEL BALL DISPOSAL SYSTEM LETS THE BALLS BE EMPTIED WITHOUT DISASSEMBLY, ALLOWING FIELD SERVICE AND MULTIPLE RUNS. STANDARD API CONNECTIONS AND HIGH-STRENGTH MATERIAL OPTIONS FOR DEEP GAS ARE AVAILABLE.

APPLICATIONS

- Wellbore cleanout
- Managed pressure drilling
- Underbalanced drilling
- Cementing operations
- Workover and completions
- Stuck pipe mitigation operations

DESIGN FEATURES

- **Single collar design:** no mid-body connections.
- **Variable cycle capability.**
- **Standard API connections.**
- **Ball drop activation:** rugged and proven.
- **Ball disposal:** balls emptied without disassembly; field serviceable, multi-run.
- **High-strength material options** for deep gas.

ITEMS TO DISCUSS

- Customizable housing for increased cycles
- Operating temperature range for the application
- Repair and maintenance options for staged repairs
- Chemical compatibility with specialist drilling fluids
- Compatibility and operation with third-party downhole tools
- Installation, handling and operating guidelines

HOW TO ORDER

Please specify: 1. tool size and top and bottom connections (standard API) 2. number of cycles required 3. material grade (high strength for deep gas, if needed) 4. operating temperature range 5. drilling fluid and chemical compatibility 6. third-party tools in the string.

A high-temperature / high-pressure version is available for demanding applications: ask for details.